Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 24

Proof by Cases

Covers Chartrand & Zhang, Section 3.4

Status of this set

Ungraded practice. Solutions released with it. Write every proof in complete sentences, and label every case.

Part I: How to Practice This Material

The template

1. **Say what the cases are** and why they cover everything: "We consider two cases, according to whether n is even or odd."
2. **Label each case** and prove it as a self-contained direct proof.
3. **Close** by noting that the cases are exhaustive, so the result holds throughout.

When to reach for cases

When the hypothesis is too weak to give you an equation. "Let n be an integer" hands you nothing; "let n be an even integer" hands you $n = 2a$.
Splitting manufactures the missing information.

Three mistakes to expect

1. **Cases that miss something.** They must cover the whole domain.
2. **Reusing a letter inside one case.** $m = 2a$ and $n = 2a$ asserts $m = n$.
3. **"Without loss of generality" on cases that are not symmetric.** Only use it when swapping names reproduces the argument exactly.

Part II: Practice Problems

Section A — One variable

1. Prove that $n^2 + n + 6$ is even for every integer n.

2. Prove that $n^2 + 5n + 2$ is even for every integer n.

Section B — Two variables

3. Let m and n be integers. Prove that $m + n$ is even if and only if m and n are of the same parity.

4. Let m and n be integers. Prove that $3n + 1$ and $5n + 2$ are of opposite parity.

Section C — Cases that are not parity

5. Prove that $|x| \geq x$ for every real number x.

6. Let A, B, C be sets. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, using cases rather than a membership table.

Section D — Writing

7. In five to eight complete sentences, explain why the cases in a proof by cases must cover the entire domain, and what goes wrong if they do not. Give a specific example of a case split that fails to cover its domain.

Section E — Challenge (optional)

8. Prove that for every integer n, exactly one of n, $n + 1$, $n + 2$ is divisible by 3.